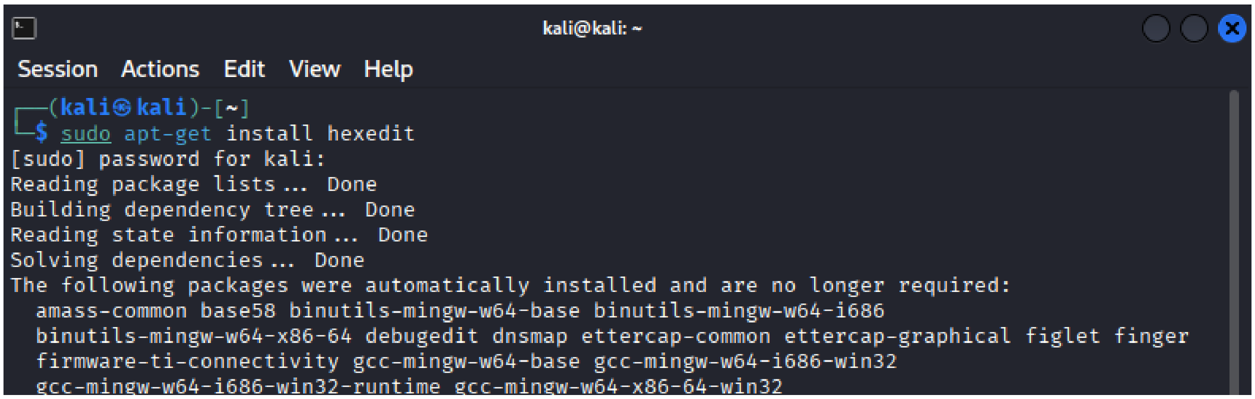
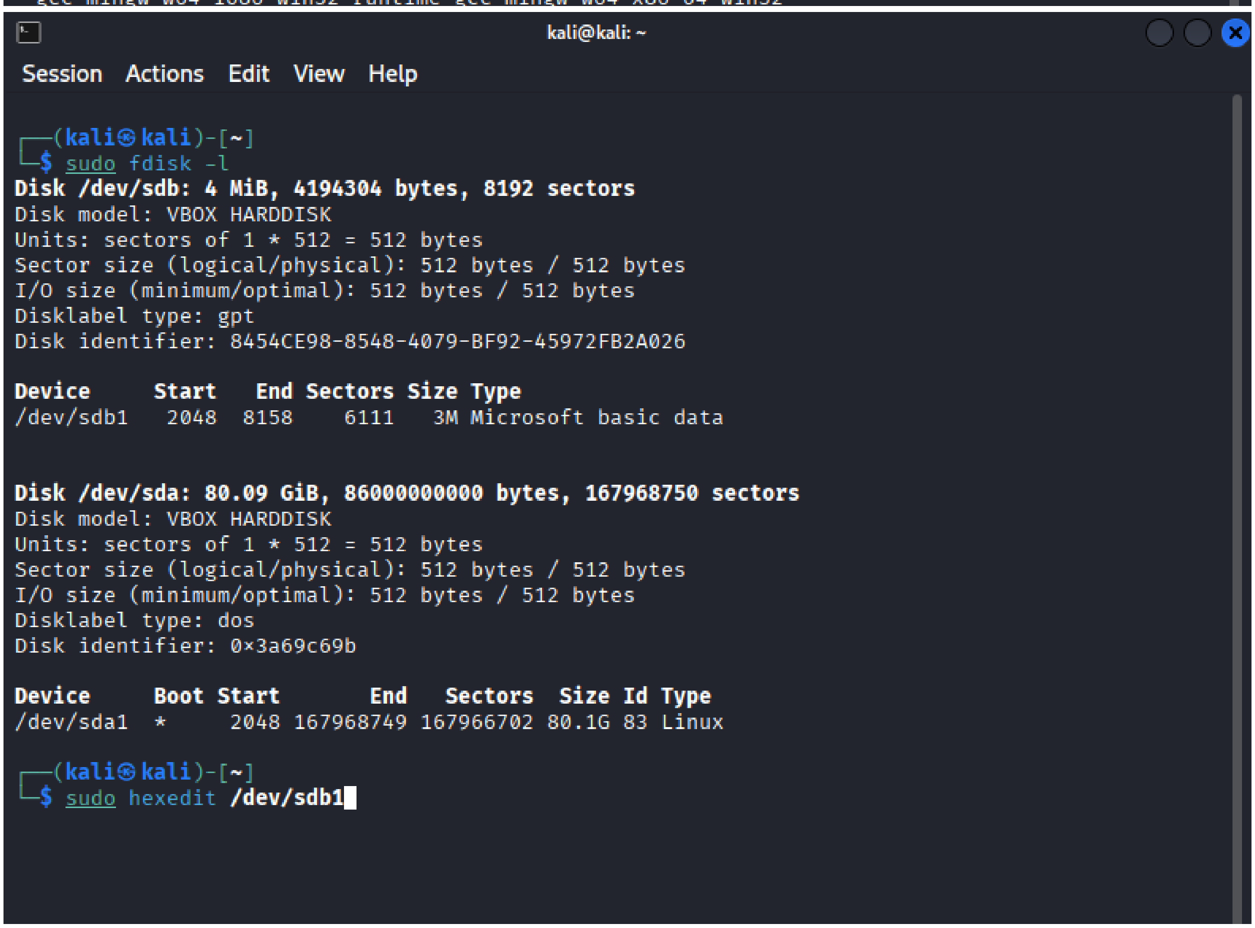
Chapter 6 Review Questions 1-19

1. Forensics software tools are grouped into **command-line** and **GUI (Graphical user interface)** applications.
2. According to ISO standard 27037, which of the following is an important factor in data acquisition? (Choose all that apply.)
    a. **The DEFR's competency**
    b. The DEFR's skills in using the command line
    c. **Use of validated tools**
    d. **Conditions at the acquisition setting**
3. An Encrypted drive is one reason to choose a logical acquisition. **True** or False?
4. Hashing, filtering, and file header analysis make up which function of digital forensics tools?
    a. **Validation and verification**
    b. Acquisition
    c. Extraction
    d. Reconstruction
5. Hardware acquisition tools typically have built-in software for data analysis. True or **False**?
6. The reconstruction function is needed for which of the following purposes? (Choose all that apply.)
    a. **Re-create a suspect drive to show what happened.**
    b. Create a copy of a drive for other investigators.
    c. Recover file headers
    d. **Re-create a drive compromised by malware.**
7. List three subfunctions of the extraction function. **Keyword Searching, Decompressing/uncompressing, and Decrypting**
8. Data can't be written to disk with a command-line tool. True or **False**?
9. Hash values are used for which of the following purposes? (Choose all that apply.)
    a. Determining file size
    b. **Filtering known good files from potentially suspicious data**
    c. Reconstructing file fragments
    d. **Validating that the original data hasn't changed**
10. In testing tools, the term "reproduceable results" means that if you work in the same lab on the same machine, you generate the same results. True or **False**?
11. The verification function does which of the following?
    a. Proves that a tool performs as intended
    b. Creates segmented files

   c. **<u>Proves that two sets of data are identical via hash values</u>**
   d. Verifies hex editors
12. What's the advantage of a write-blocking device that connects to a computer through a FireWire or USB controller? **<u>Allows investigators to easily connect suspect drives to a forensic workstation without the risk of writing data to the drive. It acts as a bridge between the suspect drive and the workstation, discarding any write commands from the OS while still allowing the OS to believe the write was successful, to preserve the integrity of the evidence.</u>**
13. Building a forensic workstation is more expensive than purchasing one. True or **<u>False</u>**?
14. A live acquisition can be replicated. True or **<u>False</u>**?
15. Which of the following is true of most drive-imaging tools? (Choose all that apply.)
    a. They perform the same function as a backup
    b. **<u>They ensure that the original drive doesn't become corrupt and damage the digital evidence</u>**
    c. **<u>They create a copy of the original drive</u>**
    d. They must be run from the command line
16. The standards for testing forensics tools are based on which criteria?
    a. U.S. Title 18
    b. ASTD 1975
    c. **<u>ISO 17025</u>**
    d. All of the above
17. A log report in forensics tools does which of the following?
    a. Tracks all file types
    b. Monitors network intrusion attempts
    c. **<u>Records an investigator's actions in examining a case</u>**
    d. Lists known good files
18. When validating the results of a forensic analysis, you should do which of the following? (Choose all that apply.)
    a. **<u>Calculate the hash value with two different tools</u>**
    b. **<u>Use a different tool to compare the results of evidence you find</u>**
    c. **<u>Repeat the steps used to obtain the digital evidence, using the same tool, and recalculate the hash value to verify the results.</u>**
    d. Use a command-line tool and then a GUI tool
19. The primary hashing algorithm the NSRL project uses is SHA-1. **<u>True</u>** or False?

Using the Hex Editor





kali@kali: ~

Session Actions Edit View Help

```
00000000   EB 3C 90 6D  6B 66 73 2E  66 61 74 00  02 04 01 00  .<.mkfs.fat.....
00000010   02 00 02 DF  17 F8 05 00  3F 00 FF 00  00 08 00 00  ........?.......
00000020   00 00 00 00  80 00 29 3D  01 DF 0B 4E  4F 20 4E 41  ......)=...NO NA
00000030   4D 45 20 20  20 20 46 41  54 31 32 20  20 20 0E 1F  ME    FAT12   ..
00000040   BE 5B 7C AC  22 C0 74 0B  56 B4 0E BB  07 00 CD 10  .□.".t.V.......
00000050   5E EB F0 32  E4 CD 16 CD  19 EB FE 54  68 69 73 20  ^..2.......This
00000060   69 73 20 6E  6F 74 20 61  20 62 6F 6F  74 61 62 6C  is not a bootabl
00000070   65 20 64 69  73 6B 2E 20  20 50 6C 65  61 73 65 20  e disk.  Please
00000080   69 6E 73 65  72 74 20 61  20 62 6F 6F  74 61 62 6C  insert a bootabl
00000090   65 20 66 6C  6F 70 70 79  20 61 6E 64  0D 0A 70 72  e floppy and..pr
000000A0   65 73 73 20  61 6E 79 20  6B 65 79 20  74 6F 20 74  ess any key to t
000000B0   72 79 20 61  67 61 69 6E  20 2E 2E 2E  20 0D 0A 00  ry again ...  ...
000000C0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000000D0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000000E0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000000F0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000100   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000110   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000120   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000130   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000140   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000150   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000160   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000170   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000180   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00000190   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000001A0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000001B0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000001C0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000001D0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000001E0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000001F0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 55 AA  ..............U.
00000200   F8 FF FF 00  40 00 05 60  00 07 80 00  09 A0 00 0B  ....@..`........
00000210   C0 00 0D E0  00 0F 00 01  11 20 01 13  40 01 15 60  ......... ..@..`
00000220   01 17 80 01  19 A0 01 1B  C0 01 1D E0  01 1F 00 02  ................
00000230   21 20 02 23  40 02 25 60  02 27 80 02  29 A0 02 2B  ! .#@.%`.'..)..+
00000240   C0 02 FF EF  02 2F 00 03  31 20 03 33  40 03 35 60  ...../..1 .3@.5`
00000250   03 37 80 03  39 A0 03 3B  C0 03 3D E0  03 3F 00 04  .7..9..;..=..?..
---  sdb1       --0x0/0x2FBE00--0%----------------------------------------------
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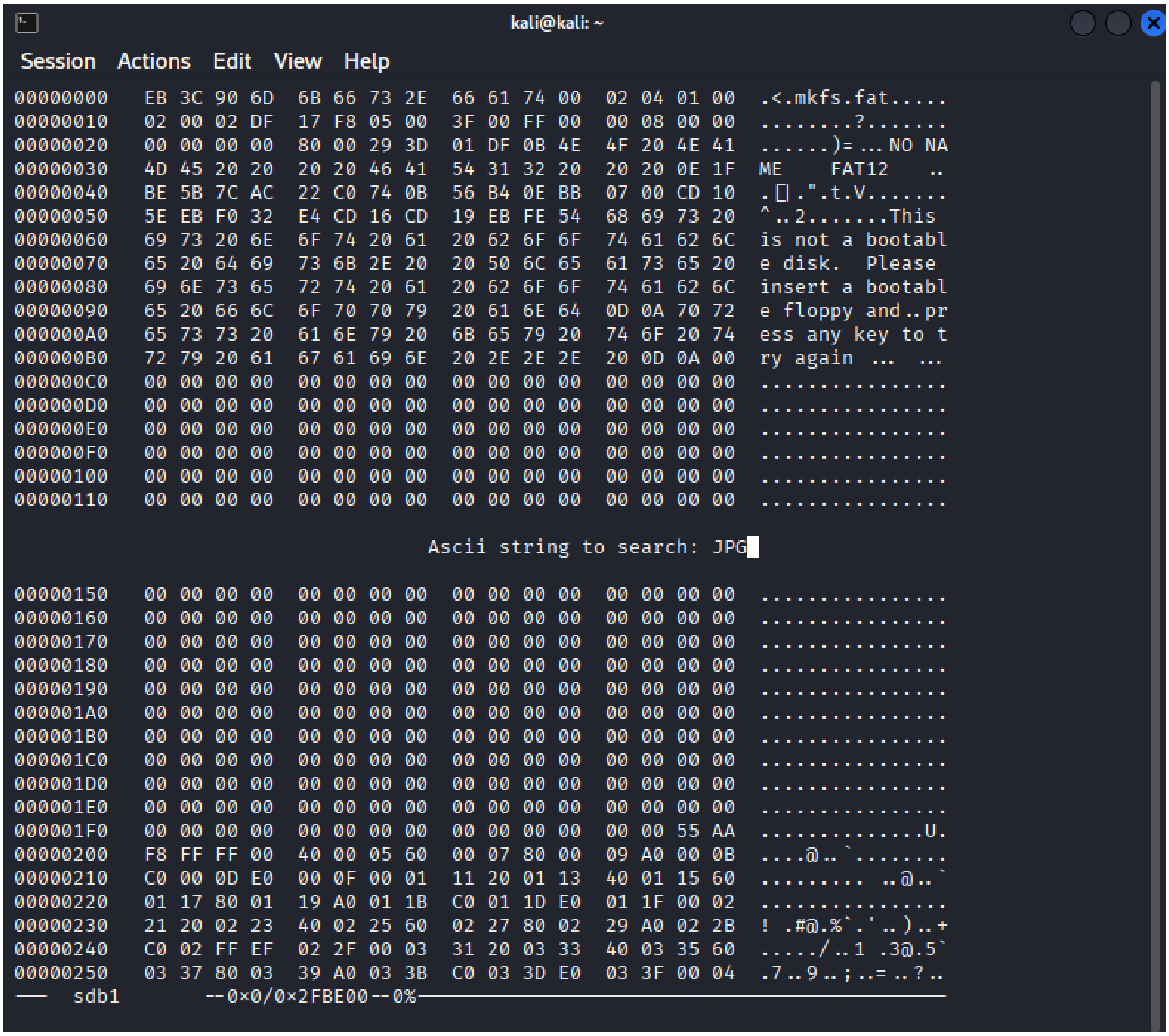
kali@kali: ~
Session Actions Edit View Help
00000000 EB 3C 90 6D 6B 66 73 2E 66 61 74 00 02 04 01 00 .<.mkfs.fat.....
00000010 02 00 02 DF 17 F8 05 00 3F 00 FF 00 00 08 00 00 ....?.......
00000020 00 00 00 00 80 00 29 3D 01 DF 0B 4E 4F 20 4E 41 ......)=...NO NA
00000030 4D 45 20 20 20 20 46 41 54 31 32 20 20 20 0E 1F ME   FAT12   ..
00000040 BE 5B 7C AC 22 C0 74 0B 56 B4 0E BB 07 00 CD 10 .[|.".t.V.......
00000050 5E EB F0 32 E4 CD 16 CD 19 EB FE 54 68 69 73 20 ^..2.......This
00000060 69 73 20 6E 6F 74 20 61 20 62 6F 6F 74 61 62 6C is not a bootabl
00000070 65 20 64 69 73 6B 2E 20 20 50 6C 65 61 73 65 20 e disk. Please
00000080 69 6E 73 65 72 74 20 61 20 62 6F 6F 74 61 62 6C insert a bootabl
00000090 65 20 66 6C 6F 70 70 79 20 61 6E 64 0D 0A 70 72 e floppy and..pr
000000A0 65 73 73 20 61 6E 79 20 6B 65 79 20 74 6F 20 74 ess any key to t
000000B0 72 79 20 61 67 61 69 6E 20 2E 2E 2E 20 0D 0A 00 ry again ... ...
000000C0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000000D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000000E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000000F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
00000100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
00000110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
Ascii string to search: JPG
00000150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
00000160 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
00000170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
00000180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
00000190 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000001A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000001B0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000001C0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000001D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000001E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
000001F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 55 AA ....U.
00000200 F8 FF FF 00 40 00 05 60 00 07 80 00 09 A0 00 0B ....@..`........
00000210 C0 00 0D E0 00 0F 00 01 11 20 01 13 40 01 15 60 ....@..`
00000220 01 17 80 01 19 A0 01 1B C0 01 1D E0 01 1F 00 02 ....
00000230 21 20 02 23 40 02 25 60 02 27 80 02 29 A0 02 2B ! .#@.%`.'..)..+
00000240 C0 02 FF EF 02 2F 00 03 31 20 03 33 40 03 35 60 ...../..1 .3@.5`
00000250 03 37 80 03 39 A0 03 3B C0 03 3D E0 03 3F 00 04 .7..9..;..=..?..
sdb1 --0x0/0x2FBE00--0%

kali@kali: ~

Session Actions Edit View Help

```
00001420   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001430   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001440   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001450   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001460   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001470   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001480   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001490   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000014A0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000014B0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000014C0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000014D0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000014E0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000014F0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001500   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001510   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001520   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001530   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001540   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001550   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001560   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001570   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001580   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001590   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000015A0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000015B0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000015C0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000015D0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000015E0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
000015F0   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  ................
00001600   42 70 00 67  00 00 00 FF  FF FF FF 0F  00 6F FF FF  Bp.g.........o..
00001610   FF FF FF FF  FF FF FF FF  FF FF 00 00  FF FF FF FF  ................
00001620   01 73 00 69  00 6E 00 67  00 6C 00 0F  00 6F 65 00  .s.i.n.g.l...oe.
00001630   74 00 6F 00  6E 00 5F 00  33 00 00 00  2E 00 6A 00  t.o.n._.3.....j.
00001640   53 49 4E 47  4C 45 7E 31  4A 50 47 20  00 00 28 87  SINGLE~1JPG ..(.
00001650   7F 46 7F 46  00 00 19 87  7F 46 F0 00  BD 43 07 00  .F.F.....F...C..
00001660   E5 45 43 54  55 52 45 32  4A 50 47 20  00 64 D0 86  .ECTURE2JPG .d..
00001670   7F 46 7F 46  00 00 D0 86  7F 46 2D 00  C7 89 01 00  .F.F.....F-.....
───   sdb1       --0×1528/0×2FBE00--0%──────────────────────────────────────
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